CONFUSION PAIRS · EXAM STRATEGY - ALL DOMAINS

Confusion pairs

Cross-cutting. Eight scenarios built from the places two things look alike and are not.

TYPE **Cross-cutting** SCENARIOS **8** TOPIC **Discrimination**

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A good distractor is not wrong nonsense. It is a true statement about the wrong thing.

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Where the default is not what people assume

THING	THE ACTUAL DEFAULT
Expectations	warn. The invalid record is written to the target and counted.
Auto Loader schema evolution	addNewColumns when no schema is supplied, none when one is.
CLONE	Deep. Shallow must be asked for explicitly.
Window frame	RANGE, not ROWS. Peers with the same ordering value group together.
Job retries	None, except on serverless and in continuous mode.
Delta streaming source	Append-only. An update in the source fails the stream.

ON THE EXAM

Every row here is a default somebody assumes is the opposite. Defaults are where the exam finds free marks.

What actually stops something happening

MECHANISM	ENFORCED?
NOT NULL and CHECK constraints	Yes. A violating write fails.
Primary key and foreign key	No. Informational, for documentation and optimisation.
Expectations	Depends on the action you chose. Warn does not stop anything.
Row filters and column masks	They restrict. They never grant.
A BROADCAST hint	Honoured even above the size threshold. That is how it kills a driver.
Runtimes below 12.2 with masks	Fail securely. No data at all is returned.

ON THE EXAM

The first two rows sit on the same table and behave oppositely. That is the single most likely confusion in Section 10.

Right technique, wrong layer

IF THE QUESTION IS ABOUT	LOOK AT
Your code's exception or print output	Driver logs, not executor logs.
A task failing during a shuffle	Executor logs, not driver logs.
Whether the cluster started at all	The cluster event log, not either of those.
An expectation failing	The pipeline event log, not the Spark UI.
Who can query a table	Unity Catalog grants, not workspace ACLs.
Who can start a cluster	Workspace ACLs, not Unity Catalog grants.

ON THE EXAM

Most wasted debugging time, and most wrong answers, come from applying a good technique at the wrong layer.

One flag, two opposite verdicts

- In module 21, **skip change commits was the trap**. Corrections needed to propagate, and the flag discarded them.
- In module 26, **the same flag is the documented answer**, for propagating a deletion.
- The difference: in module 26 the deletion is applied **explicitly to both the source and the streaming table first**.
- So the flag is right when you have already handled the change yourself, and wrong when you have not.

> ON THE EXAM
>
> The same setting, the same stream, opposite verdicts. No amount of memorising 'skipChangeCommits is bad' survives this.

Micro-batches do not overlap

- A published sample question describes a stream whose micro-batches sometimes exceed the trigger interval.
- The **correct** answer: shorten the interval so records do not back up and cause large batches that spill.
- The **distractor**: shorten it so idle executors start the next batch while earlier tasks finish.
- That is false. **A new micro-batch does not begin while the previous one is running.**

> ON THE EXAM
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> Two options recommending the same action for different reasons. Only the reasoning separates them, which is a very professional-exam thing to do.

There is no cross-task transaction

- Another published question: task A succeeds, B succeeds, C fails. What is the state?
- The published answer: A and B completed, and **some operations in C may have completed successfully**.
- **Nothing is rolled back across tasks.** A job is not a transaction.
- Which is exactly why module 30 insisted on asking whether a task is idempotent before repairing it.

> ON THE EXAM
>
> Individual Delta operations are atomic. A multi-task job is not, and two of that question's distractors offer group rollback.

Controlling file size without moving data

- A published question: write 1 TB of JSON to Parquet with 512 MB target files, **without rearranging the data**.
- The answer: set the **maximum partition bytes** setting to 512 MB, which controls partition size **on read**.
- Every distractor rearranges. **Repartition** shuffles. **Coalesce** moves data. **Sorting** repartitions.
- Output file size follows partition count, and partition count is decided when you read.

> ON THE EXAM
>
> The phrase 'without rearranging the data' is the whole question. It rules out every option that causes a shuffle.

Sometimes the objective names the wrong tool

- A published question's stated objective is about **Delta Lake clone**.
- The scenario: share a table with another team, renaming some fields and hiding others.
- **The published answer is a view**, with aliased column names and only approved fields selected.
- The question asks for the solution emphasising **simplicity**, and renaming and subsetting columns is a view's job.

> ON THE EXAM
>
> Read the requirement, not the objective heading. The objective tells you the topic area; the scenario tells you the answer.

How to use this under time pressure

- **Find the requirement sentence.** It is usually one clause, and it eliminates two options immediately.
- Ask **which layer** the question is about before choosing: data or object, driver or executor, table or pipeline.
- If two options recommend the same action, the difference is the **reasoning**. Read both fully.
- Be suspicious of an option that is true and answers a question nobody asked.

> ON THE EXAM
>
> Roughly two minutes per question. Most of that should go on understanding the requirement, not on weighing four options equally.

> A silver streaming table stopped updating. The bronze Delta table it reads from now receives corrective UPDATE statements from an operations team. An engineer sets skipChangeCommits on the streaming read and the stream runs cleanly again.
>
> A reviewer asks whether this is correct.

Is it?

A Yes. skipChangeCommits is the documented way to handle changes in a streaming source.

B No. The corrections are being discarded. The documented use requires the change to be applied to both tables first.

C No. skipChangeCommits only applies to deletes, not updates.

D Yes, provided the pipeline is fully refreshed afterwards.

◆ PAUSE AND DECIDE

Is it?

A Yes. skipChangeCommits is the documented way to handle changes in a streaming source.

It is documented for propagating deletions already applied to both tables. Not for discarding corrections.

B No. The corrections are being discarded. The documented use requires the change to be applied to both tables first.

Silver now runs cleanly while holding values the business knows are wrong.

C No. skipChangeCommits only applies to deletes, not updates.

It skips change commits generally. The scope is not the problem.

D Yes, provided the pipeline is fully refreshed afterwards.

A full refresh would reprocess, and the flag would discard the corrections again.

ON THE EXAM

The stream running cleanly is the symptom, not the cure. It is quiet because it is ignoring the thing it should be processing.

> A job has three tasks. Task A loads a staging table and succeeds. Task B enriches it and succeeds. Task C writes to an external payments API and fails halfway through.
>
> An engineer states that because task C failed, the whole run is rolled back, so it is safe to simply rerun the job from the start.

What is wrong with that reasoning?

A Nothing. Multi-task jobs are transactional across tasks.

B There is no cross-task transaction. Some operations in C may have completed, so rerunning may repeat them.

C Only Delta writes are rolled back; external API calls are not.

D Tasks A and B would also be rolled back, so rerunning is wasteful but safe.

◆ PAUSE AND DECIDE

What is wrong with that reasoning?

A Nothing. Multi-task jobs are transactional across tasks.

They are not. The guide's own published answer says some operations in the failed task may have completed.

B There is no cross-task transaction. Some operations in C may have completed, so rerunning may repeat them.

A job is not a transaction, and an external payments call is not idempotent.

C Only Delta writes are rolled back; external API calls are not.

Even the Delta writes in A and B are not rolled back. They committed.

D Tasks A and B would also be rolled back, so rerunning is wasteful but safe.

They completed and committed. Nothing undoes them.

ON THE EXAM

C is the sophisticated wrong answer: it half-knows the rule and invents a boundary where Delta writes roll back and external calls do not.

> A wide managed table has 90 columns. Analysts filter on a column at position 78 and complain of full scans. An engineer recalls that Delta collects statistics on the first 32 columns and proposes reordering the schema so the column moves into the first 32.
>
> The table is Unity Catalog managed with predictive optimization enabled.

What should they be told?

A. The reordering is correct; 32 columns is the universal limit.

B. The 32-column limit applies to external tables. Managed tables under predictive optimization have no such limit, so something else explains the scans.

C. Reorder the schema and also run OPTIMIZE to rebuild statistics.

D. Statistics cannot be collected on managed tables at all.

◆ PAUSE AND DECIDE

What should they be told?

A The reordering is correct; 32 columns is the universal limit.

It is not universal. That is the update most material has missed.

B The 32-column limit applies to external tables. Managed tables under predictive optimization have no such limit, so something else explains the scans.

Statistics columns are chosen intelligently on managed tables. The diagnosis needs to start again.

C Reorder the schema and also run OPTIMIZE to rebuild statistics.

Two actions built on a premise that does not apply here.

D Statistics cannot be collected on managed tables at all.

They are collected, and chosen more intelligently than on external tables.

ON THE EXAM

The engineer's fact is correct and its scope is not. The honest answer is that the diagnosis has to restart, which is often the right answer.

> A compliance officer asks how long deleted records remain recoverable. An engineer answers 30 days, citing Delta's default history retention. A second engineer answers 7 days, citing vacuum.
>
> The tables are managed with predictive optimization enabled and default settings throughout.

Who is right?

A The first. History retention governs recoverability, and it defaults to 30 days.

B The second, in practice. Vacuum removes the underlying files after 7 days, and once they are gone no version can be read.

C Neither. Deleted records are unrecoverable immediately.

D Both are wrong; deletion vectors keep records indefinitely.

◆ PAUSE AND DECIDE

Who is right?

A The first. History retention governs recoverability, and it defaults to 30 days.

History retention governs the log. It cannot make files readable after vacuum has removed them.

B The second, in practice. Vacuum removes the underlying files after 7 days, and once they are gone no version can be read.

Two properties, and the files are the binding constraint. Predictive optimization runs vacuum automatically.

C Neither. Deleted records are unrecoverable immediately.

With deletion vectors the row is not even rewritten, let alone removed.

D Both are wrong; deletion vectors keep records indefinitely.

Only until purge and vacuum, which predictive optimization performs.

ON THE EXAM

Both engineers cite a real number. Only one of them is the binding constraint, and it is the one governing the files.

> A job must write a large dataset out with a specific target file size. A colleague proposes a repartition before writing. Another proposes coalesce. A third proposes sorting the data, which repartitions as a side effect.
>
> The requirement states explicitly that the data must not be rearranged.

Which approach meets the requirement?

A Repartition to the calculated number of partitions before writing.

B Set the maximum partition bytes configuration to the target size, then read, transform and write.

C Coalesce to the calculated number of partitions before writing.

D Set the shuffle partitions configuration and sort the data before writing.

◆ PAUSE AND DECIDE

Which approach meets the requirement?

A Repartition to the calculated number of partitions before writing.

Repartition shuffles, which is rearranging.

B Set the maximum partition bytes configuration to the target size, then read, transform and write.

Controls partition size on read, so output files match without any shuffle. The published answer.

C Coalesce to the calculated number of partitions before writing.

Moves data between partitions. Cheaper than a repartition, still rearranging.

D Set the shuffle partitions configuration and sort the data before writing.

Sorting repartitions as a side effect, which the option itself admits.

ON THE EXAM

One clause decides everything. Three options are eliminated by 'must not be rearranged' before you evaluate their arithmetic.

> An analyst has SELECT on a table but cannot query it. A colleague suggests the table's row filter is blocking them and should be removed. Another suggests adding a column mask so they see something rather than nothing.
>
> The analyst receives a permissions error, not an empty result.

What is actually needed?

A Remove the row filter, which is denying access.

B USE CATALOG and USE SCHEMA on the parents. Filters and masks restrict; they never grant.

C Add a column mask so the analyst sees masked values instead of an error.

D Grant the analyst ownership of the table.

◆ PAUSE AND DECIDE

What is actually needed?

A Remove the row filter, which is denying access.

A row filter returns fewer rows. It does not produce a permissions error.

B USE CATALOG and USE SCHEMA on the parents. Filters and masks restrict; they never grant.

A permissions error means no access at all, and SELECT alone is never sufficient.

C Add a column mask so the analyst sees masked values instead of an error.

Masks restrict what a permitted user sees. They cannot create permission.

D Grant the analyst ownership of the table.

Enormous overreach, and ownership is a single principal.

ON THE EXAM

The error type is the discriminator. A permissions error means no access; an empty result would mean access plus a filter.

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> A long-running job succeeded after a repair. A duration warning fired. Separately, a Structured Streaming job on the same cluster is showing one task far slower than the median with lopsided shuffle read, and spill on two tasks.
>
> A single engineer must explain both.

What are the two explanations?

A Both are memory pressure; the cluster needs resizing.

B Duration spans every repair attempt including idle time; and a few slow tasks with skewed shuffle read is data skew, not memory pressure.

C Both are skew; the repair took long because of the same skewed data.

D The duration warning is a false positive and the spill indicates memory pressure across the cluster.

◆ PAUSE AND DECIDE

What are the two explanations?

A Both are memory pressure; the cluster needs resizing.

Neither is. One is duration accounting, one is skew.

B Duration spans every repair attempt including idle time; and a few slow tasks with skewed shuffle read is data skew, not memory pressure.

Two unrelated symptoms with two documented explanations, and spill on two tasks is a consequence of skew.

C Both are skew; the repair took long because of the same skewed data.

Invents a connection between two unrelated jobs.

D The duration warning is a false positive and the spill indicates memory pressure across the cluster.

The warning is correct by its own definition, and spill on two tasks is localised.

ON THE EXAM

Two symptoms, two causes, no connection. The temptation is to find one explanation for everything on one cluster.

> A team must share a curated table with another department. The field names differ from the other department's conventions, and three columns are not approved for them to see.
>
> The recipient works in the same Databricks workspace. The team asks whether to use a deep clone, a shallow clone, or Delta Sharing.

What is the simplest solution?

A. A deep clone into a table with the required schema, synchronised on a schedule.

B. A view selecting only approved fields, with the column names aliased to their conventions.

C. A shallow clone, which is instant and references the same files.

D. Delta Sharing, which is designed for sharing between teams.

◆ PAUSE AND DECIDE

What is the simplest solution?

A. A deep clone into a table with the required schema, synchronised on a schedule.
A clone cannot rename or drop columns, and this adds a copy to keep in sync.

B. A view selecting only approved fields, with the column names aliased to their conventions.
Renaming and subsetting columns is exactly a view's job, with no copy and nothing to synchronise. The guide's published answer.

C. A shallow clone, which is instant and references the same files.
Also cannot rename or drop columns, and exposes everything.

D. Delta Sharing, which is designed for sharing between teams.
Built for sharing outside your metastore. The recipient is in the same workspace.

ON THE EXAM

Three of the four options are copying mechanisms answering a question about presentation. The requirement is rename and subset.